

Innovation 2026

QData

Conversational AI for Secure &
Governed Enterprise Data Access



Production Ready



Enterprise Secure



Multi-Agent AI

The Problem

Data is locked behind SQL expertise.

Business Users Blocked

No SQL expertise means every data request goes through a data engineer, no matter how simple.

Engineers Bottlenecked

Data teams spend hours on repetitive ad-hoc queries instead of high-value engineering work.

BI Tools Fall Short

Existing tools require training, have rigid interfaces, and lack intelligent multi-step reasoning.

There is no lightweight, secure, and governable solution that lets users ask data questions in plain English and get accurate, audited answers — without exposing raw infrastructure or compromising data access policies.

The Solution

"Type a question. Get the answer. No SQL. No waiting."

User "Show me the top 10 customers by revenue this quarter"

QData Understands intent → Identifies schema → Executes SQL securely → Returns answer

"The top 10 customers by revenue this quarter are: Acme Corp (\$2.4M), GlobalTech (\$1.9M)..."

Plain English Input

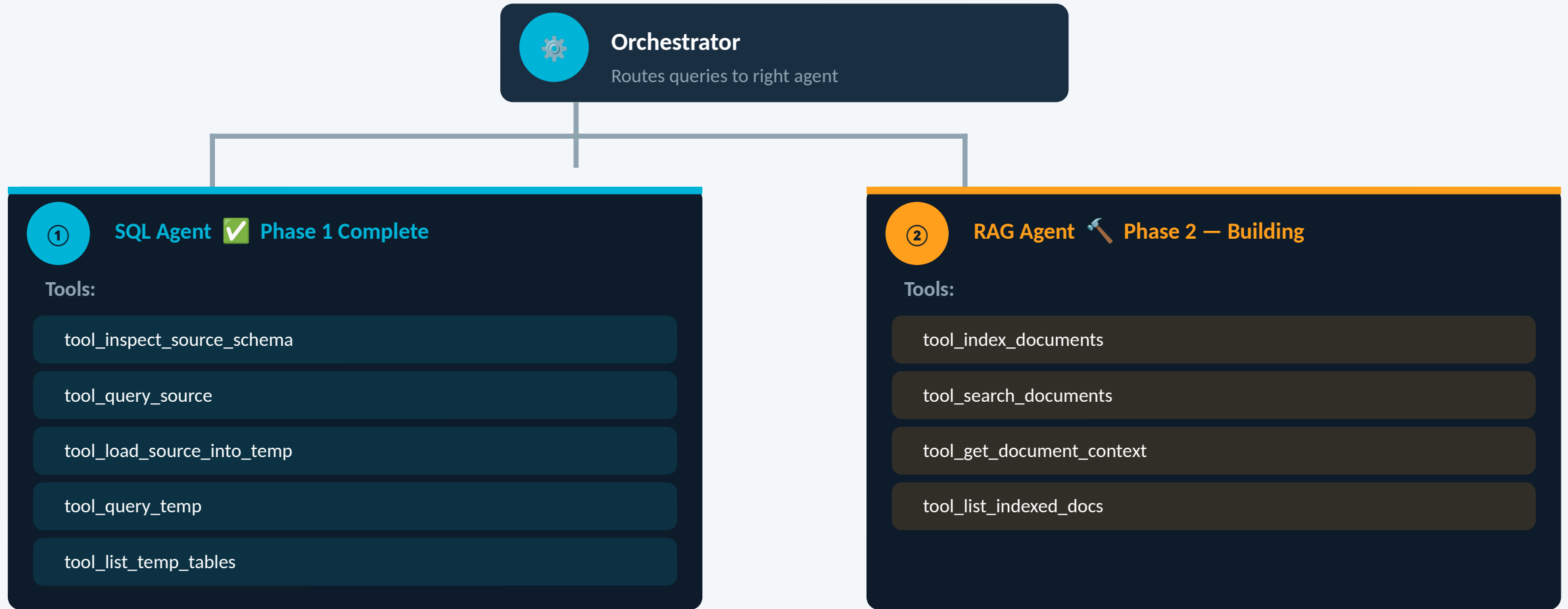
Auto SQL Generation

Governed Access

Full Audit Trail

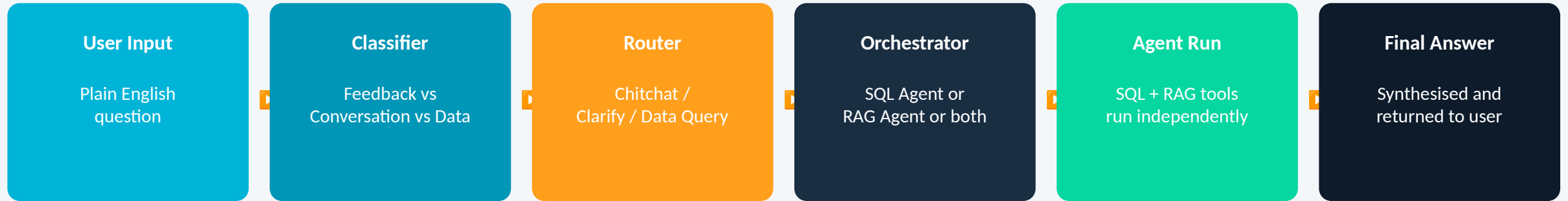
Built on a fully custom multi-agent architecture — SQL Agent + RAG Agent coordinated by an Orchestrator.

Multi-Agent Architecture

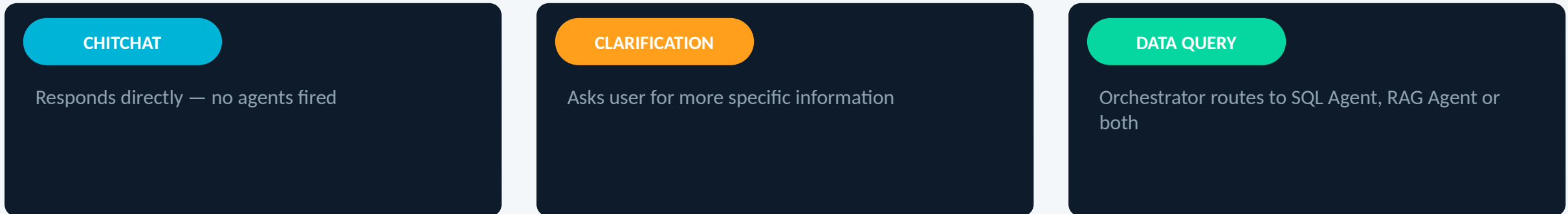


Each agent has its own loop • own scratchpad • own tools • own LangSmith trace

How It Works



Router decides the path:



Enterprise Security & Governance



AST-Level SQL Validation

SQL parsed into syntax tree — INSERT, UPDATE, DELETE, DROP blocked at parse time, not by keywords.



JWT Authentication

Every MCP request requires signed JWT. HS256 local secret for dev, AWS KMS RS256 for production.



Redshift SSO

Queries run as the actual user. Redshift natively enforces table and row-level permissions.



Full Audit Trail

Every run, every tool call logged in LangSmith with session ID, tools used, and termination reason.



Session Isolation

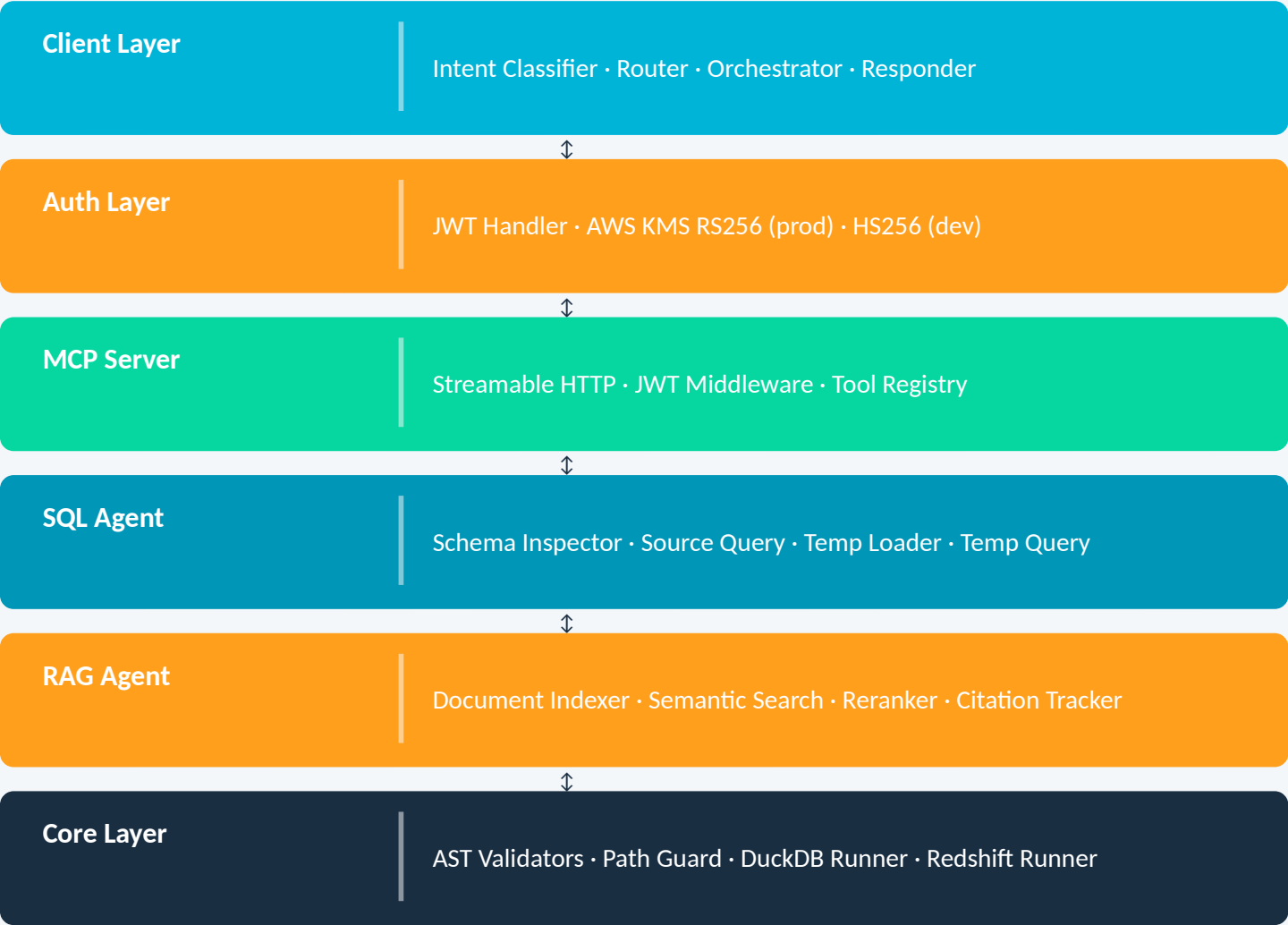
Each conversation gets its own isolated DuckDB sandbox. Temp data wiped automatically on session end.



Structured Output

LLM forced to return Pydantic-validated schema. No prompt injection via malformed JSON possible.

Technical Architecture



Dev vs Production		
Feature	Dev	Prod
Data Source	CSV+DuckDB	Redshift
Auth Signing	HS256	KMS RS256
Permissions	Path Guard	Redshift SSO
SQL Agent	✓	✓
RAG Agent	✓	✓
AST Valid.	✓	✓
LangSmith	✓	✓
Switch	ENV=dev	ENV=prod

Agent Intelligence

Live Conversation

- You:** hi
CHITCHAT → responds directly
- Bot:** Hello! How can I help you today?
- You:** show top 5 customers by revenue
DATA_QUERY → SQL Agent
- Bot:** The top 5 customers are: Acme (\$2.4M)...
- You:** summarise the Q3 report
DATA_QUERY → RAG Agent
- Bot:** Q3 report highlights: Revenue up 12%...
- You:** great thanks
POSITIVE_FEEDBACK → logged
- Bot:** Glad that helped!

Agent Step-by-Step

Step 1

tool_inspect_source_schema()

← Schema: sales → customers, orders

Step 2

tool_query_source(sql)

← SELECT ... FROM sales.customers...

Step 3

tool_search_documents(query)

← Q3 report: revenue up 12%...

Final

Orchestrator synthesises both results

← Combined answer returned to user

Observability & Audit Trail

LangSmith Trace — Every Run

Run: qdataagent-turn

chain

SQL Agent run

agent

tool_inspect_source_schema

tool

tool_query_source

tool

RAG Agent run

agent

tool_search_documents

tool

Final Answer

output

1

Per-Run Logging

Every turn logged with session ID, tools used, steps count, and termination reason.

2

Child Run Tracing

Each tool call logged as a child run — full drill-down visibility per agent.

3

User Feedback Scoring

Positive scores 1.0, negative 0.0 — tied to run_id for complete traceability.

4

Token Usage Tracking

Input, output, total token counts per run for cost monitoring and optimisation.

Business Impact



Seconds

Ad-hoc request
resolution vs days



Zero

SQL expertise
needed by users



100%

Queries audited
and traceable



Any

Client warehouse
deployable instantly

Reusability Across AIA Data Vertical

QData requires minimal configuration change per client — any organisation running Amazon Redshift, Snowflake, or a structured data warehouse can deploy this multi-agent system and immediately enable natural language data access for their teams.

Current Status & Roadmap

Phase 1 — SQL Agent Complete

-  Fully working terminal-based agent
-  Amazon Redshift + CSV connectivity
-  Multi-schema dot notation queries
-  AST-level SQL mutation prevention
-  JWT auth (HS256 dev / KMS RS256 prod)
-  LangSmith audit trail — every run traced
-  Production-grade intent router
-  Structured output — zero hallucinated calls
-  Session isolation and temp sandbox
-  MCP server — tools as HTTP service

Roadmap — Next Phases

Phase 2

RAG Agent

PDF ingestion, semantic chunking, hybrid vector+BM25 search, reranking, citation tracking. Connects as second agent to orchestrator.

Phase 3

Web UI

React chat interface for end users. File upload for CSV and PDFs. REST API via FastAPI app server.

Phase 4

Docker + AWS ECS

Containerised deployment. docker-compose for local. Scalable cloud deployment on ECS.

Phase 5

Multi-Agent Orchestration

Parallel SQL + RAG execution. Cross-domain synthesis. Complex multi-hop analytical workflows.

QData

Conversational AI for Secure & Governed Enterprise Data Access

2

Agents
(SQL + RAG)

6+

Security
Layers

100%

Production
Ready